Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia[1]

E. IRVING
Pacific Geoscience Centre, P.O. Box 6000, Sidney, B.C., Canada V8L 4B2

G. J. WOODSWORTH
Geological Survey of Canada, 100 West Pender St., Vancouver, B.C., Canada V6B 1R8

P. J. WYNNE
Pacific Geoscience Centre, P.O. Box 6000, Sidney, B.C., Canada V8L 4B2

AND

A. MORRISON
Department of Chemistry, University of Victoria, Victoria, B.C., Canada V8W 2L2

Received June 11, 1984
Revision accepted October 29, 1984

The mid-Cretaceous Spuzzum and Porteau plutons of the Coast Plutonic Complex of British Columbia have two magnetizations, *A* and *B*. The *A* magnetization (eight sites, 83 specimens, $D = 30.3°$, $I = 56.7°$, $\alpha_{95} = 4.9°$, paleolatitude = 37 ± 5°N, paleopole 65.0°N, 14.9°W, $A_{95} = 6.2°$) is considered to have been acquired in the age range 105–90 Ma. This result differs from the field established for cratonic North America in this time range. The difference could be caused either by previously undetected tilting about a horizontal axis of the plutons, or by their rotation about a vertical axis and lateral displacement relative to the craton. Previously observed mid-Cretaceous magnetizations from other rock units from the western Canadian Cordillera and the Cascades of Washington, United States, are similarly discordant with respect to the craton. This similarity over such a large area indicates that, although local undetected tilting could be partly responsible, it is unlikely to be the prime cause, and we argue therefore that lateral displacement and rotation have occurred. It would seem that much of the western part of the Canadian Cordillera has moved north by about 2400 km and rotated clockwise since the mid-Cretaceous. The paleolatitude of the southern Coast Plutonic Complex of British Columbia is statistically identical to that recently observed (39 ± 3°N) for three plutons from the Central Sierra Nevada of California, which raises the possibility that the two complexes were much closer together at the time of their emplacement than at present. The second magnetization called *B* (four sites, 27 specimens, $D = 5.1°$, $I = 67.6°$, $\alpha_{95} = 4.7°$, paleopole 86.5°N, 51.2°W) is parallel to the mid-Tertiary field, as previously determined from nearby intrusions, and is considered to be an overprint acquired during regional heating and low-grade metasomatism. Some earlier paleomagnetic studies of mid-Cretaceous rocks from the Coast Plutonic Complex indicated either an absence of displacement or uncertain evidence for it, and we attribute this to the nonrecognition, in this earlier work, of similar magnetically stable overprints of Tertiary age. Overprints in several Triassic rock units in the western Cordillera are parallel to the *A* magnetization, indicating that the mid-Cretaceous and the mid-Tertiary probably were periods of severe magnetic overprinting in British Columbia. Mid-Cretaceous and Late Triassic results from the western Cordillera of British Columbia are systematically different, indicating that movements relative to the craton occurred between these times.

Les plutons de Spuzzum et de Porteau d'âge crétacé moyen du Complexe plutonique côtier de la Colombie-Britannique exhibent deux aimantations, *A* et *B*. L'aimantation *A* (huit sites, 83 spécimens, $D = 30{,}3°$, $I = 56{,}7°$, $\alpha_{95} = 4{,}9°$, paléolatitude = 37 ± 5°N, paléopole 65,0°N, 14,9°O, $A_{95} = 6{,}2°$) semble avoir été acquise entre 105 et 90 Ma. Ce résultat diffère du champ établi pour le craton de l'Amérique du Nord relatif à cet épisode de temps. Cette différence peut s'expliquer soit par une inclinaison non décelée antérieurement autour de l'axe horizontal des plutons, ou par une rotation autour d'un axe vertical accompagnée d'un déplacement latéral relativement au craton. Les aimantations observées antérieurement d'âge crétacé moyen provenant de d'autres unités lithologiques de la Cordillère canadienne occidentale et des Cascades de Washington, les États-Unis, sont semblablement discordantes par rapport au craton. Cette ressemblance couvrant une si grande région révèle que, même si l'inclinaison locale non décelée peut en être partiellement la cause, il est peu probable que ce soit la raison principale, et nous proposons plutôt un déplacement et une rotation. Il apparaîtrait qu'un large portion de la Cordillère canadienne occidentale ait migré vers le nord d'une distance approximative de 2400 km et qu'elle ait subi une rotation dans le sens des aiguilles d'une montre depuis le Crétacé moyen. La palèolatitude de la partie sud du Complexe plutonique côtier de la Colombie-Britannique est statistiquement identique à celle récemment observée (39 ± 3°N) pour trois plutons de la Sierra Névada du centre de la Californie, ce qui suggère la possibilité que les deux complexes étaient beaucoup proches l'un de l'autre au moment de leur mise en place qu'ils le sont actuellement. La deuxième aimantation appelée *B* (quatre sites, 27 spécimens, $D = 5{,}1°$, $I = 67{,}6°$, $\alpha_{95} = 4{,}7°$, paléopole 86,5°N, 51,2°O) est parallèle au champ du Tertiaire moyen, telle que déterminée pour les intrusions avoisinantes, et elle est interprétée comme une surimpression acquise lors d'une période de réchauffement régional et de métasomatisme de faible degré. Des études paléomagnétiques plus anciennes des roches d'âge crétacé moyen du Complexe plutonique côtier ont suggéré soit un absence de déplacement ou une incertitude à le démontrer, et nous attribuons ceci au défaut d'avoir observé, dans ces études antérieures, des surimpressions semblables, magnétiquement stables, d'âge tertiaire. Les surimpressions de plusieurs unités lithologiques triassiques de la Cordillère occidentale sont parallèles à l'aimantation *A*, révélant que le Crétacé moyen et le Tertiaire moyen furent des périodes de surimpressions magnétiques intenses en Colombie-Britannique. Les résultats obtenus pour le Crétacé moyen et le Trias supérieur pour la Cordillère occidentale de la Colombie-Britannique sont systématiquement différents, indiquant que des mouvements relatifs au craton sont apparus entre ces périodes de temps.

[Traduit par le journal]

Can. J. Earth Sci. 22, 584–598 (1985)

[1]Contribution of the Earth Physics Branch No. 1134.

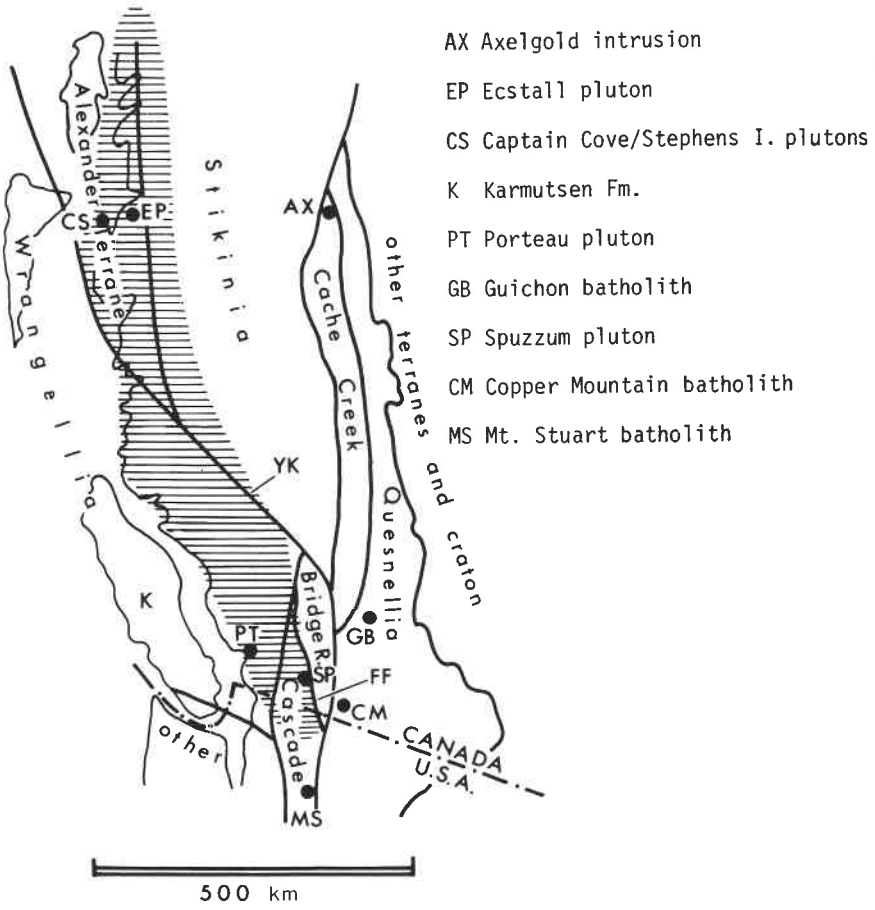


FIG. 1. General locality map. Shown are the main terranes, the paleomagnetic sampling localities of Cretaceous rocks, and the older rocks from which overprints of probable Cretaceous age have been obtained (see Table 3). YF and FF are the Yalokum and Fraser fault systems. The Coast Plutonic Complex is horizontally shaded.

Introduction

Paleomagnetic directions of Triassic and Jurassic rocks of western and central British Columbia diverge strongly from reference paleofields for cratonic North America. The divergence indicates that much of British Columbia was then situated more than 1000 km south of its present position relative to the craton. These observations were obtained from Stikinia (Monger and Irving 1980) and Wrangellia (Irving and Yole 1972: Schwarz *et al.* 1980; Yole and Irving 1980), and most were derived from rocks whose structural attitudes are well known so that the inclination of the paleofield could be determined and the paleolatitude accurately calculated. Results from some intrusions, whose structural attitudes are less certain, also yielded southerly paleolatitudes (Monger and Irving 1980; Symons 1983*a*). Of these the Cretaceous Axelgold intrusion (AX of Fig. 1) is of interest because it indicated that the northward motion occurred after the Early Cretaceous Epoch. However, Axelgold is not very accurately dated, and there is a problem in determining paleohorizontal at the time its magnetization was acquired. These uncertainties do not invalidate the arguments for displacement from southerly latitudes, but they make the magnitude and timing of the displacements uncertain (Monger and Irving 1980). In contrast, previous observations from Cretaceous intrusions of the Coast Plutonic Complex (shaded in Fig. 1), situated between Stikinia and Wrangellia, were inconclusive. Some indicated little or no displacement (Symons 1973*b*). Others, which did diverge from the cratonic paleofield, were explained by local tilting rather than by displacement (Symons 1977*a*,*b*). Despite these considerable uncertainties, Beck *et al.* (1981*a*) have speculated that, like Stikinia and Wrangellia, the major mid-Cretaceous plutons of the western edge of North America, including the Coast Plutonic Complex of British Columbia, originated farther south. In the present paper we describe new results and analyses from the southern part of the Coast Plutonic Complex, which broadly confirm the speculations of Beck *et al.* (1981*a*).

An important requirement of paleomagnetic studies is some measure of the paleohorizontal at the time magnetizations were acquired. In well bedded units attitudes can be easily determined, but these are not always available and one is obliged to study intrusive bodies, as we have done in this study. We contend that in large plutons, sampled well away from faults or other known structural complications, the present horizontal is a reasonable approximation to paleohorizontal, because they were probably magnetized during uplift and cooling, and because uplift in such massive bodies is likely to have occurred vertically without appreciable tilting. This assumption is justified if uniform directions of magnetization can be observed among several coeval plutons spaced over a wide area, because if tilting had occurred it is unlikely to have been the same everywhere.

Geology and sampling

We have sampled two bodies from the southern Coast Plu-

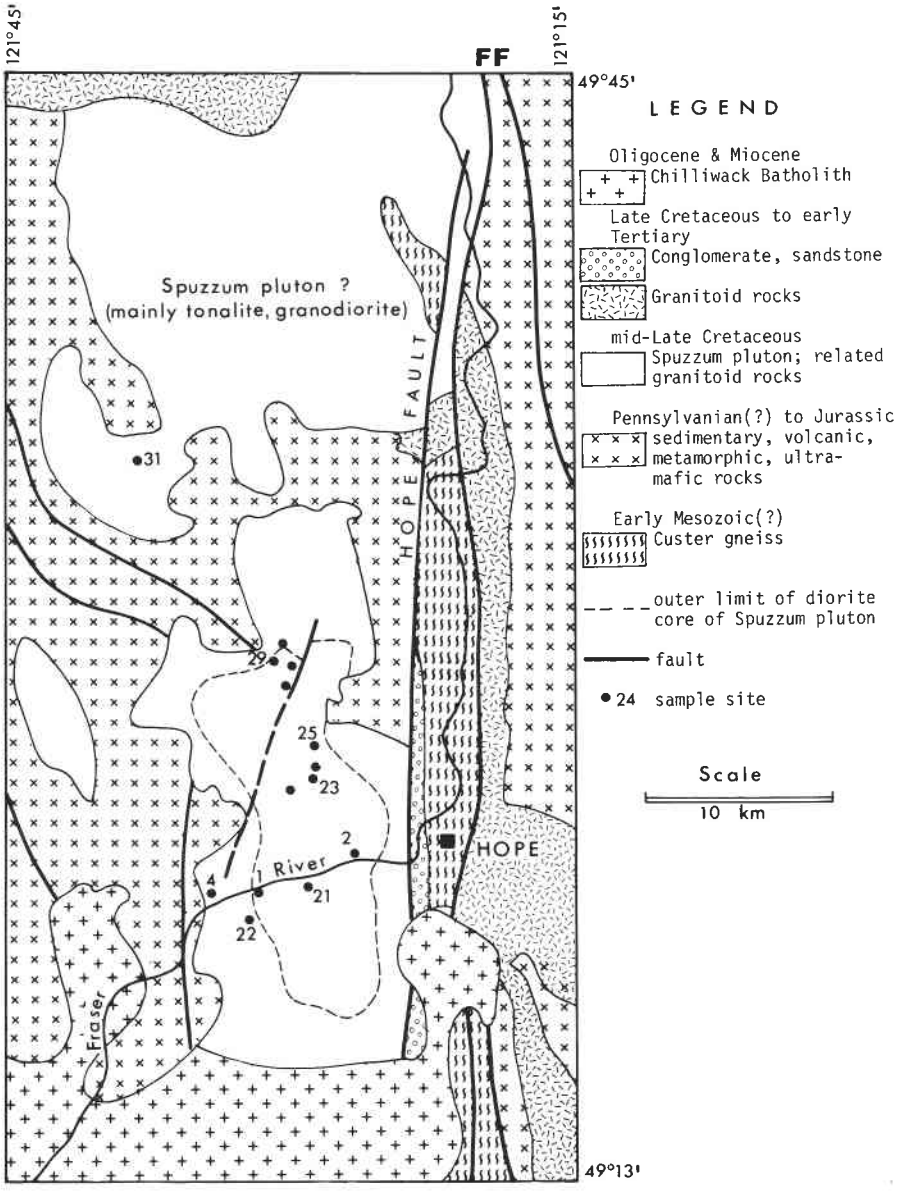


FIG. 2. Sampling sites in the Spuzzum pluton. Geology adapted from Monger (1970), Richards and McTaggart (1976), Vining (1977), and Bartholomew (1979). Sampling sites correspond to those in Table 1. Sites (24, 26, 27, 28, 30) with incoherent magnetization are not numbered. The locations (°N latitude, °W longitude) of sampling sites are as follows: (1) 49°21.7′, 121°33.9′; (2) 49°22.7′, 121°29.6′; (4) 49°21.6′, 121°35.9′; (21) 49°21.9′, 121°31.7′; (22) 49°21.0′, 121°34.4′; (23) 49°24.9′, 121°31.3′; (24) 49°24.5′, 121°32.3′; (25) 49°25.7′, 121°31.3′; (26) 49°25.1′, 121°31.2′; (27) 49°28.3′, 121°32.4′; (28) 49°27.8′, 121°32.4′; (29) 49°28.4′, 121°33.1′; (30) 49°28.6′, 121°32.9′; (31) 49°34.0′, 121°39.1′.

tonic Complex, the Spuzzum and Porteau plutons, and some associated sedimentary rocks of the Gambier Group (SP and PT in Fig. 1). The Spuzzum pluton, near the town of Hope, intrudes a belt of schists and gneisses that is an extension, into British Columbia, of the core zone of the northern Cascade terrane of Washington (Monger 1970). The pluton lies between Harrison Lake and the Fraser River and is situated west of the Yalokum and Fraser River fault systems (YK and FF in Fig. 1).

The Spuzzum pluton is a granitoid body 50 km long by 10 km wide (Richards and McTaggart 1976; Vining 1977). It is split by an east–west belt of metamorphic rocks (Bartholomew 1979) into two bodies, which are roughly similar in composition, structure, and age. All but one of our sample sites are in the more mafic, better studied southern half (Fig. 2). The southern body is irregularly zoned and consists of a central dioritic complex rimmed by tonalite. The core consists of hypersthene–augite diorite and hornblende diorite. The northern body is mainly tonalite and granodiorite that grade locally into diorite.

The Spuzzum pluton was intruded either during the waning stages of regional metamorphism of the surrounding schists or shortly afterwards (Richards and McTaggart 1976; Gabites 1983). Uplift of the area began in the mid-Cretaceous and was accompanied by high-angle normal faulting (Bartholomew 1979). On the east, early Tertiary faulting (the Fraser River fault system, of which the Hope Fault is the most westerly strand) created a graben into which Eocene(?) sandstones and conglomerates were shed (Fig. 2). These strata are tilted between 40 and 90° to the west (Monger 1970). The Spuzzum pluton, the Fraser River fault system, and the Eocene sediments

are cut by the Chilliwack composite batholith, the various parts of which are dated between 40 and 16 Ma (Richards and McTaggart 1976). The intrusion of the Chilliwack batholith and related plutons was the last important thermal event in the area. They have been studied paleomagnetically by Symons (1973*a*) and by Beck *et al.* (1982).

Five K/Ar dates on hornblende from the southern body of the Spuzzum pluton range from 81 to 90 Ma, and five on biotite range from 79 to 103 Ma (Richards and McTaggart 1976; McLeod *et al.* 1976). Aside from two biotite dates of 103 ± 5 and 101 ± 3 Ma, all K/Ar dates fall between 79 and 90 Ma. The K/Ar ages young towards the Hope Fault. In the centre and west of the body all K/Ar dates are 90 Ma or older. A discordant $^{238}U/^{206}Pb$ date on zircons from near our site 21 is 110 ± 3 Ma (R. L. Armstrong, personal communication, 1984) and this is taken to be the maximum age of emplacement.

The northern body is less well dated than the southern. K/Ar dates range from 76 ± 4 Ma, obtained on a concordant biotite–hornblende pair, to 92 ± 3 Ma, obtained from hornblende (Bartholomew 1979; McTaggart and Thompson 1967). Gabites (1983, and personal communication, 1984) obtained an Rb–Sr mineral isochron of 77 ± 4 Ma and a $^{238}U/^{206}Pb$ date on zircons of 125 Ma from diorites near our site 31. These data suggest emplacement in Early Cretaceous time, and setting or resetting of the K–Ar and Rb–Sr systems later in the Cretaceous Period.

The region containing the Porteau pluton is separated from the Cascade Terrane, which contains the Spuzzum pluton, by a fault along Harrison Lake (Fig. 1). The Porteau pluton is mostly tonalite, with subordinate granodiorite and diorite (Fig. 3). It intrudes Lower to Middle Cretaceous volcanics and sediments of the Gambier Group, which is thought by Woodsworth and Tipper (1980) to have been deposited on the basement of Wrangellia. The Gambier Group is folded and has been metamorphosed by the intrusion of the Porteau pluton. R. L. Armstrong (personal communication, 1983) has obtained a preliminary U/Pb date on zircons of 100 ± 2 Ma from a sample collected very near our sampling site 8. This is probably a maximum age for emplacement. No K/Ar dates are available from the Porteau, but generally similar plutons to the north and south give dates between 90 and 117 Ma. We have also sampled three metamorphosed argillite beds of the Gambier Group because it was thought that they might record the cooling field of the pluton. Earlier, Symons (1973*b*) sampled quartz diorites and granodiorites from various plutons (apparently including both the Porteau pluton and the Squamish granodiorite noted below) over a distance of about 80 km on the eastern shore of Howe Sound and farther north in the Cheakamus River valley. He argued that his results are concordant with others from cratonic North America. His results are considered later.

Geological evidence concerning possible tilting of the plutons is derived from two sources. The K/Ar ages record progressive cooling of the Spuzzum pluton from west to east and could constitute indirect evidence of tilting. Bartholomew (1979) thought that the ages signified a westerly tilting through an unknown angle during movement along the Fraser River fault system, towards which uplift and erosion increased. Alternatively, the pluton may have been uplifted without tilting from west to east, along a series of steep faults parallel to the Hope Fault. More direct evidence of tilting in the region is that the Burrard and Kitsilano formations, which overlie the Coast Plutonic Complex in the lower Fraser Valley to the south and west of the sampling area, have dips of about 10° to the south. The beds are Late Cretaceous or early Tertiary in age. The centre of the basin in which these sedimentary rocks were deposited lay to the south, so some of the dip could be original. To what extent these dips apply in a general way to the Spuzzum or Porteau plutons is not known. Clearly the geological evidence regarding possible tilting of the plutons is not definitive.

Study of trial samples collected by J. W. H. Monger and R. L. Armstrong from the Squamish granodiorite at the north of Fig. 3 indicated that such acidic material does not possess a stable remanent magnetization. Initial directions were very scattered, with generally downward inclinations. Treatment in alternating fields (AF) yielded a very rapid fall in intensity and no improvement in grouping. In contrast, trial specimens of diorite from the Spuzzum pluton generally (but not always) yielded excellent data, as described below, and subsequently, therefore, we always sampled the most basic rocks available.

One hundred and two cores (from which about 200 specimens were cut and studied) were collected from 15 sites. In the southern Spuzzum pluton, nine sites are from diorite and three from tonalite. No site is closer than 3 km to the Hope Fault, and none shows evidence of shearing or later hydrothermal alteration.

Description of magnetizations

For the most part the magnetizations fall into two general groups, one in which magnetizations are systematically directed and one in which they are not. The distinction can be made at two levels: by the grouping of directions amongst specimens from the same site, and by the coherence of magnetization within a specimen.

We use Schonstedt fluxgate magnetometers, which provide 12 readings, from which four independent determinations of the magnetization vector are obtained. A useful measure of within-specimen homogeneity (magnitude of nondipole components) can be obtained by calculating the precision, k_s, of the four determinations (Fisher 1953). The values of k_s are of the order of 10^2 to occasionally 10^3 for homogeneous specimens, in which the alignment of the directions of magnetization of individual particles is good, and of 10 or less for specimens in which it approaches a random distribution.

Four sites have incoherent magnetization (their locations are plotted in Figs. 2 and 3 and listed in the captions). The average k_s is 5, and directions in specimens from the same site are very scattered. During AF or thermal demagnetization, directions change randomly and neither k_s nor the within-site scatter (k_w) is improved. Evidently, no memory of the paleofield has been preserved.

Directions at sites 27 and 28 were not random, but AF and thermal demagnetization failed to resolve any systematically directed magnetization. They apparently contain multicomponent magnetizations whose stability fields overlap to such an extent that no separation could be made.

At collecting sites with coherent magnetization, two types of behaviour can be recognized, *A* and *B*, and there is some variability within each type. Ideal single component magnetizations (referred to as *A1*) were observed at eight sites. They are directed east of north, with downward inclinations mainly between 50 and 60°. Unblocking temperatures are limited to a small high-temperature interval (575–650°C in the example shown in Fig. 4, ranging up to 680°C in other specimens). AF demagnetization yields an excellent linear decay of

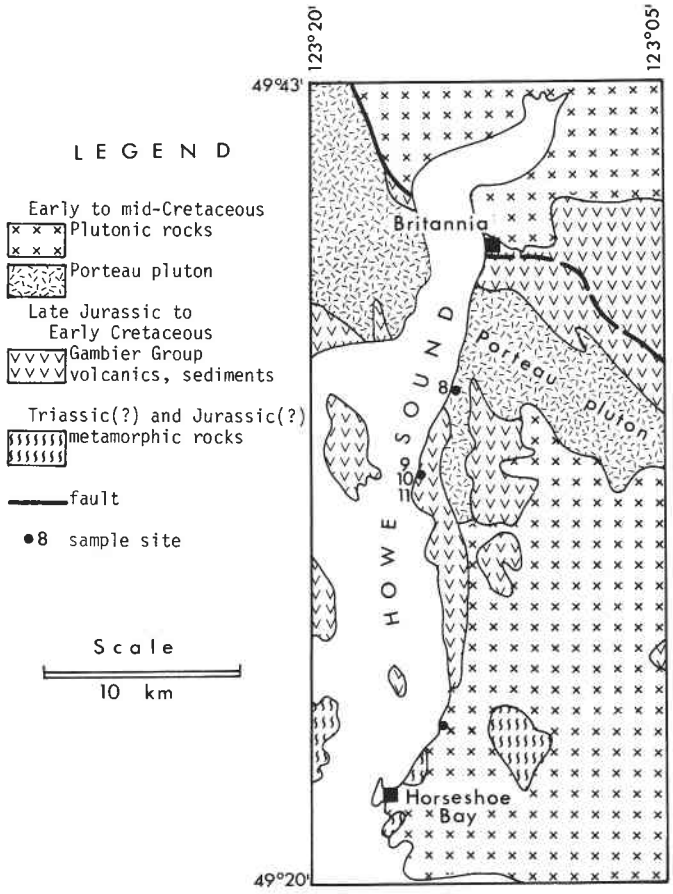


FIG. 3. Sampling sites in the Porteau pluton. Geology after Roddick and Woodsworth (1979). Sampling sites correspond to those in Table 1. Site 10 has incoherent magnetization. The locations (°N latitude, °W longitude) of sampling sites are as follows: (8) 49°33.8′, 123°13.9′; (9), (10), and (11) 49°31.3′, 123°15.5′.

intensity to the origin on orthogonal plots (Fig. 5). Although the intensity is much diminished, the directions are unchanged after treatment in 100 mT. The k_s values in *A1* specimens typically range from 150 to 300. Under reflected light, minute intergrowths of opaque minerals were observed in mafic minerals too small for optical identification, and these are probably the carriers of the very hard magnetization, as Hargraves and Young (1969) and Evans and McElhinny (1969) have shown in other basic intrusive rocks. The *A1* directions, averaged by sites, are listed in Table 1 and plotted individually in Fig. 6. The overall averages, giving sites unit weight, are listed in Table 2. Entry 1 is for the Spuzzum pluton only. Entry 2 includes results from the Porteau pluton.

The *A2* magnetizations are less perfect. They have roughly the same directions as *A1*, but are more scattered (Fig. 6). Unblocking temperatures range up to 650°C, and orthogonal plots of thermal and AF demagnetization decay to the origin but in an irregular fashion (Fig. 7). During demagnetization the directions vary 10 or 20° in an apparently random manner. The k_s values average about 20, much lower than for *A1* specimens. Only at site 22 does *A2* behaviour occur.

Another subordinate form of behaviour (*A3*) occurs at site 23 (Fig. 8). During AF demagnetization, the directions migrate generally from northeast toward the south without reaching an end point. Orthogonal plots are linear but do not march to the origin, so an unresolved magnetization, with a remanent

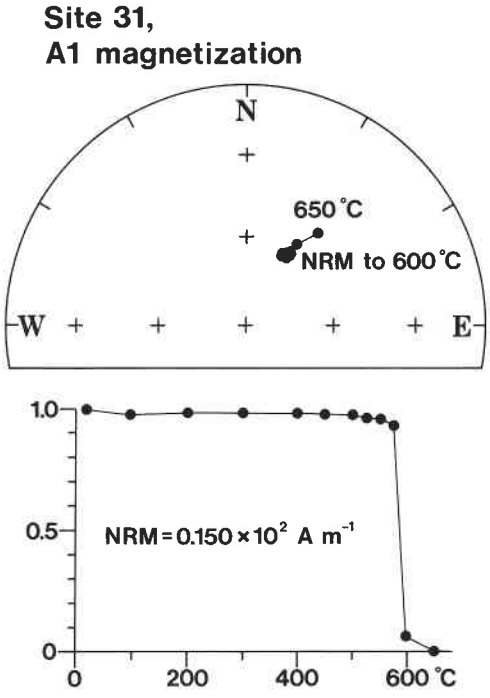


FIG. 4. Thermal demagnetization of *A1* magnetization in hornblende diorite at site 31, Spuzzum pluton. Direction changes above, intensity changes below.

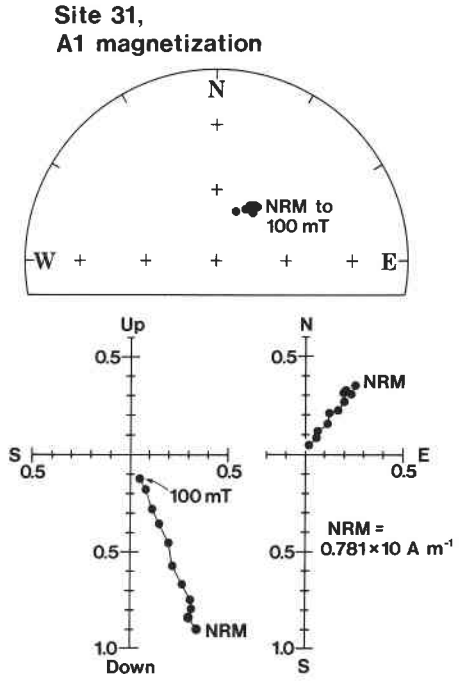


FIG. 5. Alternating field demagnetization of *A1* magnetization in hornblende diorite at site 31, Spuzzum pluton. Direction changes above, orthogonal plots below.

coercive force exceeding 100 mT, is present. The removed magnetization (labelled RM in Fig. 8), has a lower but still high remanent coercive force, and, although results from different specimens are somewhat scattered, their mean points to the north-northeast with downward inclinations, approximately parallel to the *A1* magnetizations (Fig. 6).

B magnetizations have directions that are northerly rather

TABLE 1. Results by sites

Site (mag.)	Rock	$N(S)$	M (A/m)	$D°$, $I°$	k_w	$\alpha_{95}°$	Treatment
01(*A1*)	H.d.	06(10)	0.1	026, 62	77	06	50 mT, 500°C
02(*A1*)	H.d.	10(20)	0.2	036, 57	154	03	50 mT, 500°C
04(*B1*)	Ton.	03(04)	0.2×10^{-2}	227,−64	14	26	10−50 mT, 200−300°C
04(*A1*)	Ton.	03(06)	0.4×10^{-2}	022; 50	96	07	10−90 mT, 525°C
08(*A1*)	Ton.	05(10)	0.1	023, 54	107	05	50 mT, 525°C
09(*B1*)	Arg.	05(10)	0.2×10^{-2}	359, 65	110	05	20−90 mT, 100−300°C
11(*B1*)	Arg.	02(04)	0.2×10^{-2}	353, 60	49	13	30−70 mT, 100−300°C
21(*A1*)	H.d.	07(14)	0.3	028, 58	97	04	50 mT, 525°C
22(*A2*)	H.d.	05(10)	0.4×10^{-1}	359, 42	18	12	20−90 mT, 525°C
23(*A3*)	H.d.	04(07)	0.7×10^{-1}	035, 46	13	17	RM 10−80 mT
23(*A1*)	H.d.	04(04)	0.7×10^{-1}	051, 62	134	08	525°C
25(*B2*)	H.d.	02(09)	0.1×10^{-1}	359, 73	79	06	50 mT, 450−650°C
29(*A1*)	H.d.	04(07)	0.1	027, 49	188	04	50 mT, 550°C
31(*A1*)	H.d.	06(12)	0.2×10^{2}	035, 60	283	03	50 mT, 550°C

NOTES: Sites are identified by accession numbers, which have no geological significance. The various magnetizations (mag.) are explained in the text. Rock types are: H.d., hornblende diorite; Ton., tonalite; Arg., argillite. $N(S)$ is the number of cores (specimens). M is the NRM intensity. D, I are the declination and inclination of the mean with respect to present horizontal; K_w is the within-site precision (Fisher 1953); and α_{95} is the radius of the circle of confidence ($P = 0.05$). The treatment used to isolate the magnetizations present is given in the last column. At site 23, RM indicates the magnetization removed between treatments stated.

TABLE 2. Summary of results

		Paleofield			Paleopole		
	$B(S)$	$D°$, $I°$	k_B	$\alpha_{95}°$	Lat. °N, Long. °W.	K	A_{95}
1. Spuzzum (*A1* magnetization)	07*(73)	031.4, 57.1	120	5.5	64.6, 016.9	76	7.0
2. Spuzzum (*A1*) and Porteau (*A1*)	08*(83)	030.3, 56.7	129	4.9	65.0, 014.9	81	6.2
3. Spuzzum (*A1*−*3*) + Porteau (*A1*)	10*(100)	026.9, 54.6	59	6.3	65.6, 007.6	40	7.7
4. Spuzzum *B1* and *B2* overprints	02 (13*)	017.1, 71.5	26	8.3	—	—	—
5. Porteau argillite (*B1* overprint)	02 (14*)	357.5, 63.5	84	4.4	—	—	—
6. Spuzzum and Porteau overprints	04 (27*)	005.1, 67.6	35	4.7	86.5, 051.2	—	—
7. Cascade Tertiary intrusions (3*)	13 (120)	002.2, 67.3	1150	3.6	88.2, 068.3	—	5.1
8. Cascade Tertiary intrusions (4*)	18 (142)	005.8, 69.4	230	6.1	84.6, 081.6	—	9.4
9. Mid-Tertiary of North America (12*)	—	—	—	—	86.7, 126.8	203	3.1

NOTES: $B(S)$ is the number of sampling sites (specimen). D, I are the declination and inclination of the mean with respect to present horizontal; k_B is the between-site precision, α_{95} is the radius of the circle of confidence; K and A_{95} are the precision of site poles and radius of the circle of confidence around the paleopole ($P = 0.05$). Unit weight is given to sites or specimens as indicated by asterisks. Results from mid-Tertiary intrusions of the Cascade Mountains are given in entries 7 and 8. Entry 7 is an average of results from those intrusions that have yielded results of high accuracy, with errors (α_{95}) ranging from 1.5 to 3.6° (Chilliwack batholith, 30−36 Ma (Beck *et al.* 1982); Grotto and Snoqualmie batholiths, 27−15 Ma (Beske *et al.* 1973); and Hope Complex 41−38 Ma (Symons 1973*a*)). In entry 8 a result of lower accuracy is also included (Mount Barr Complex, 16−21 Ma, $\alpha_{95} = 8°$, Symons 1973*a*). Entry 9 is an average of 12 paleopoles from North America ranging in age from 15 to 39 Ma and listed in Irving and Irving (1982, Table 1). Note the good agreement of entries 6−9.

than northeasterly and are more steeply inclined than *A* (Fig. 6). Usually (*B1* magnetization) the unblocking temperatures are low, but occasionally (*B2*) they are high. *B1* occurs typically in argillites of the Gambier Group associated with the Porteau pluton (sites 9, 11, Table 1, Fig. 3). The *B1* always has high remanent coercive force: the directions and intensity remain little changed in AF fields up to 100 mT (Fig. 9). The unblocking temperatures are predominantly below 400°C (Fig. 10), so that much of the *B1* magnetization was acquired at relatively low temperatures. The magnetizations, however, persist above 500°C, although poorly grouped. Values of k_s for the *B1* group average about 80. The *in situ* directions of magnetization at both argillite collecting sites are in excellent agreement (Table 1), but after restoration to paleohorizontal, the directions ((*D*, *I*) 349°, 31° (site 9) and 248° 61° (site 11)) differ. Hence *B1* magnetization in the argillites postdates its deformation.

B1 also occurs alongside *A1* magnetization in the tonalite of the Spuzzum pluton at site 4. Some specimens possess only an *A1* magnetization, but others contain both *A1* and *B1* (Fig. 11). In this example *B1* is stronger than *A1*, but both are weak (10^{-3} A/m) compared with those for igneous rocks generally. The natural remanent magnetization (NRM) is directed initially towards the south, with steep upward inclination. During AF demagnetization the directions change in an irregular way and the intensity increases, showing that an essentially antiparallel magnetization (an *A1* magnetization) is being removed. The removed magnetization is shown in Fig. 11 by a star labelled RV (20−40 mT) and has a direction 27°, + 67°. During thermal demagnetization the directions remain little changed up to

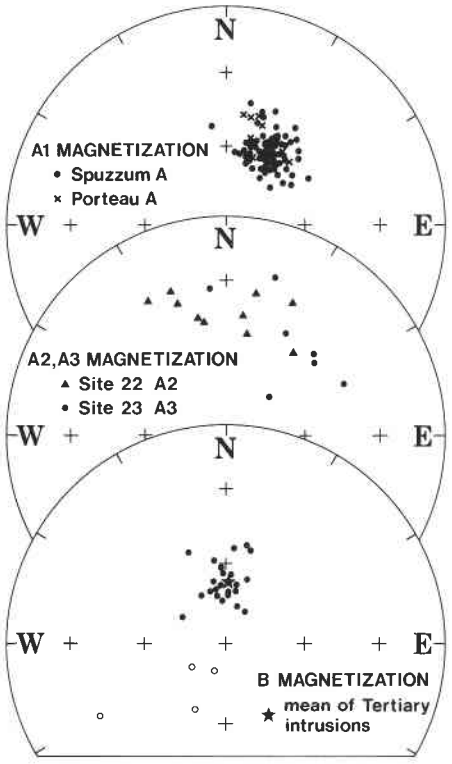


FIG. 6. Directions of magnetization components observed. Top, *A1* magnetization. Middle, *A2* and *A3* magnetizations. Bottom, *B* magnetizations; the mean direction of magnetization of mid-Tertiary intrusions (Table 2, entry 7) of northern Cascade Mountains shown for comparison. Solid (open) symbols denote downward (upward) inclinations.

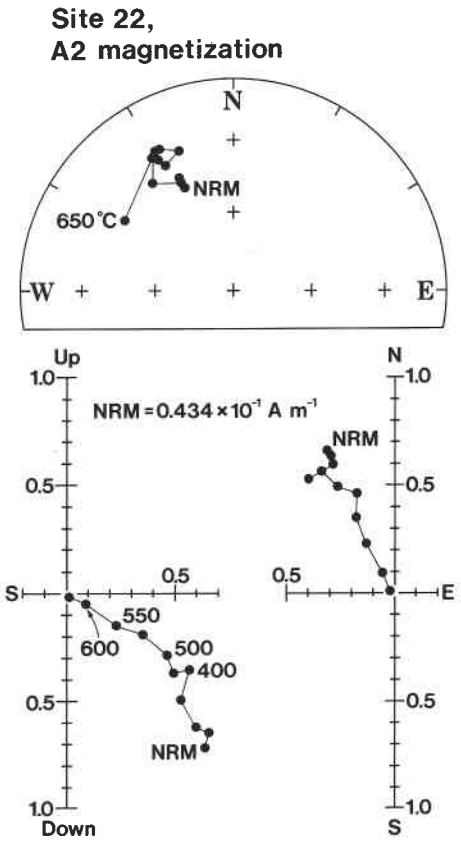


Fig. 7. Thermal demagnetization of *A2* magnetization in hornblende diorite at site 22, Spuzzum pluton. Direction changes above, orthogonal plots below.

300°C and then move quickly to a typical *A1* direction with a strong decrease in intensity as the *B1* magnetization (RV 300−400°C) is removed. In this instance the directions of the removed *A1* and *B1* magnetizations are essentially antiparallel, but this is not common. At site 4 most estimates of *B1* have a steeper inclination than *A1*. This is shown by the end-point estimates of *A1* and *B1* in Fig. 11 (high AF yield an inclination for *B1* of about −70°, and thermal demagnetization at high temperature yields an inclination of about 55° for *A1*) and from the average values of *A1* and *B1* at site 4 (Table 1).

B2 magnetizations with high unblocking temperatures (between 550 and 650°C) are rare, occurring only in diorite at site 25 (Fig. 12). The k_s values average about 60.

Origins of magnetizations

The *A1* magnetizations are simple univectorial magnetizations. The within-specimen coherence (k_s) is high, the directions in specimens from the same site and from different sites are well grouped, and remanent coercivities are high. Unblocking temperatures are always high (550−675°C), much higher than that of radiogenic argon in biotite (300°C), and are comparable to that in hornblende (550°C), but lower than blocking temperatures for U/Pb in zircons (approximately 800°C). It needs to be emphasized that all samples were obtained outside the zone of K/Ar resetting; the Spuzzum samples were taken at a distance of more than 3 km from the Fraser River fault system in the region where K/Ar ages exceeded 90 Ma. Hence the age of the *A1* magnetization in the southern part of the Spuzzum pluton is probably 90 Ma or more. At site 21 the well defined *A1* magnetization (Table 1) almost certainly formed between 110 and 101 Ma, the U/Pb zircon and biotite K/Ar dates at that site, probably, we suggest, nearer the lower than the upper limit. By the same argument, the age of the *A1* magnetization from site 31 in the northern body of Spuzzum lies between 77 Ma (the Rb−Sr mineral isochron) and 125 Ma (the U/Pb zircon date) and probably close to 92 Ma, the hornblende K/Ar date nearest to site 31. Similarly the age of the *A1* magnetization in the Porteau pluton is probably between 90 Ma (the youngest K/Ar dates on nearby biotites) and 110 Ma (the U/Pb zircon date). Taken altogether, the *A1* magnetizations of Spuzzum and Porteau probably were acquired during the interval 105−90 Ma, which falls within the Cretaceous Normal Superchron (113−83 Ma, Harland *et al.* 1982). *A1* magnetizations are always normally polarized, consistent with this conclusion.

The *A2* and *A3* magnetizations, although more scattered, have essentially the same properties; apparently they comprise an *A1* magnetization with, in *A2*, superimposed random noise and, in *A3*, systematic (and separable) noise. The *A1*, *A2*, and *A3* magnetizations do not differ in direction significantly (Tables 1 and 2), and there is no reason to believe that they differ in age. The inclusion of the *A2* and *A3* magnetizations in the grand average (entry 3, Table 2) contributes an undesirable element of random noise (between-site precision k decreases from 129 to 59, Table 2). Hence the best estimate of the mean paleofield is given in entry 2 of Table 2. The directions (site 8 in Table 1 and crosses of the upper part of Fig. 6) from the coeval Porteau pluton are in excellent agreement with those from Spuzzum, indicating that the two bodies had approxi-

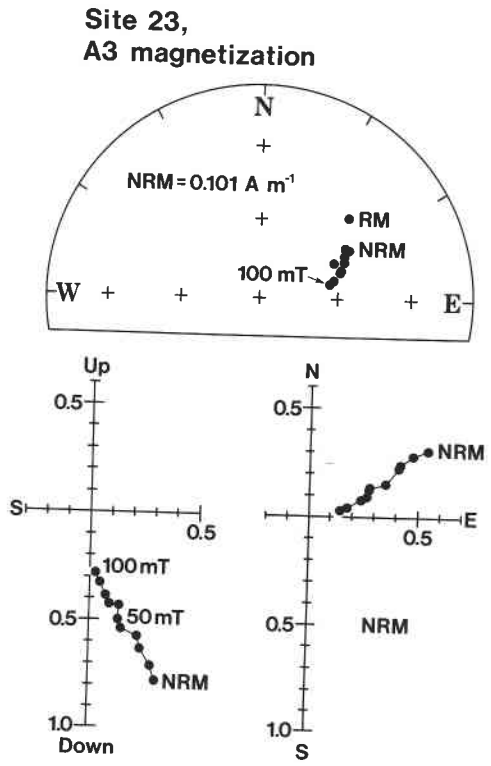


FIG. 8. Removal of *A3* magnetization (labelled RM) in hornblende diorite at site 23, Spuzzum pluton, by AF demagnetization. Direction changes above, orthogonal plots below.

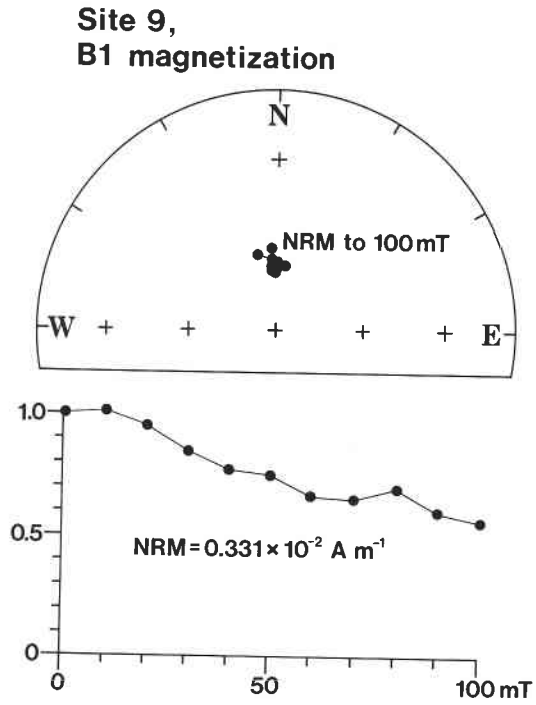


FIG. 9. AF demagnetization of *B1* magnetization in argillite at site 9, marginal to Porteau pluton. Direction changes above, intensity changes below.

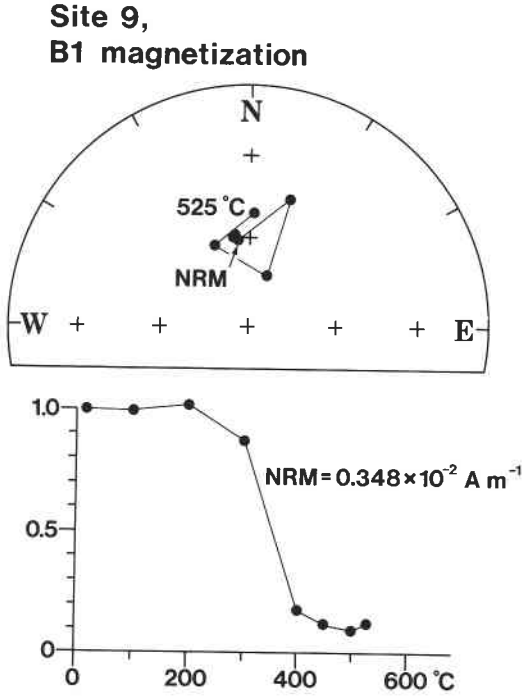


FIG. 10. Thermal demagnetization of *B1* magnetization in argillite at site 9. Direction changes above, intensity changes below.

mately the same relative positions at the time of cooling and have not moved appreciably relative to one another since then.

B magnetizations occur in all rock types sampled. *B1* occurs in Spuzzum tonalite and Porteau argillite, *B2* in Spuzzum diorite. *B1* is commonly marked by low unblocking temperature and high remanent coercive force, typical of pyrrhotite. Very fine grained pyrrhotite was tentatively identified in polished sections of Porteau argillite. Larger sulphide grains are also present and appear to be pyrite and chalcopyrite. Sulphide grains occur in polished sections from other rock specimens with *B* magnetization, but they are too small to be identified optically. There are a few reversed magnetizations, but normal polarity is the most common (Fig. 6). The mean *B* direction is in excellent agreement with the mean direction observed in the mid-Tertiary instrusions of the northern Cascades in the region of the sampling area (Fig. 6 and Table 2). Hence the *B1* magnetization, because it is characterized by low unblocking temperatures, probably was acquired at the time of cooling following a period of mild heating caused by the intrusion of these mid-Tertiary bodies. The rare *B2* magnetization, with its high unblocking temperatures, was presumably acquired by localized metasomatism affecting the iron minerals, induced perhaps by circulation of hydrothermal solutions during the same intrusive episode.

Displacements or tilts?

The paleopole calculated from the *A1* magnetization of the Spuzzum and Porteau plutons is listed in Table 2 and plotted in Fig. 13. It is assumed in these calculations that the plutons have not been appreciably tilted, that is, that the present horizontal is a good approximation to the paleohorizontal at the time the *A1* magnetization was acquired. The paleopole differs greatly in position from that expected from cratonic North America. There are two ways to interpret this difference: either both plutons could have remained where they are now relative to cratonic North America and have been tilted uniformly, or they could have been displaced latitudinally and rotated about a vertical axis relative to the craton. To explain the result by tilting requires that both the Spuzzum and Porteau plutons were tilted 29° in a direction 237° east of north (to the southwest).

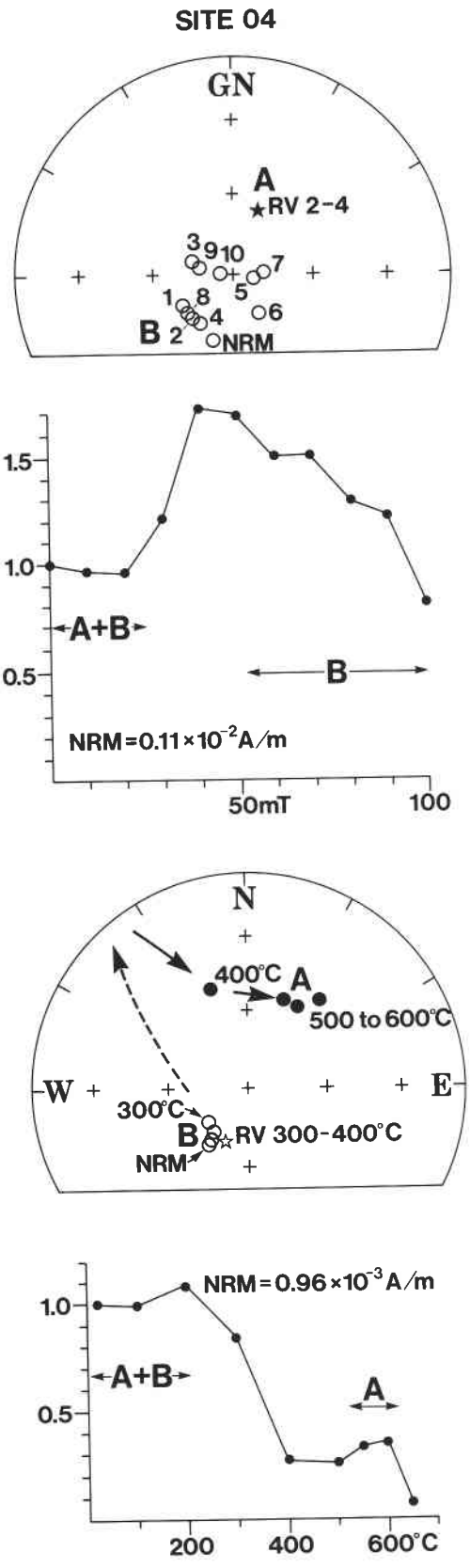


FIG. 11. *A1* and *B1* magnetizations occurring in the same tonalite specimen at site 4. Direction (points numbered in tens of milliteslas) and intensity changes during af demagnetization above, same for thermal demagnetization below. Removed magnetizations (RV) denoted by stars. Open (solid) symbols are upward (downward) inclinations. Directions plotted with respect to present horizontal.

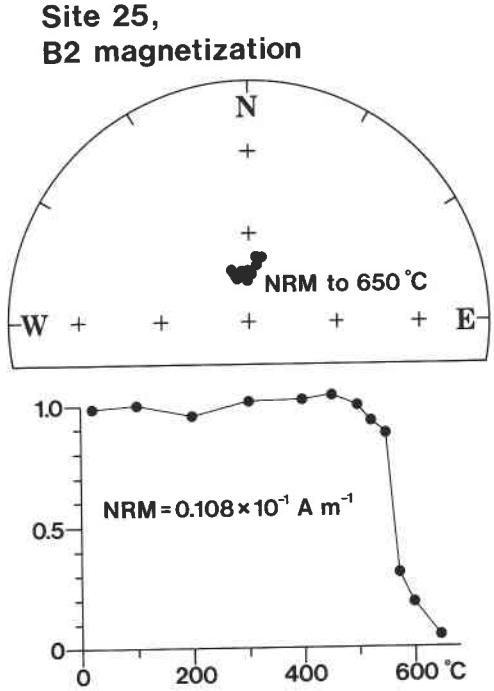


FIG. 12. Thermal demagnetization of *B2* magnetization in hornblende diorite at site 25, Spuzzum pluton. Direction changes above, intensity changes below.

Such a tilt could be a composite of the westerly tilt invoked by Bartholomew (1979) from the evidence of K/Ar ages and a southerly tilt evidenced by the onlap of the Burrard and Kitsilano formations. As already noted, neither of these evidences is compelling and the choice between tilt or lateral motion depends ultimately on the regional consistency of paleomagnetic data and on comparisons with results from elsewhere in the Coast Plutonic Complex. These comparisons are now made and we wish to emphasize at the outset that the data presently available are insufficient to resolve this dilemma.

Paleopoles from other mid-Cretaceous rock units from the Cascade Mountains of Washington and the Canadian Cordillera are plotted in Fig. 14. All have been obtained from magnetizations that can be reasonably assigned to mid-Cretaceous time. All are normally magnetized, as expected, since they fall in the Cretaceous Normal Superchron. We have not included several results from earlier sampling of mid-Cretaceous plutons of the Coast Plutonic Complex for the reasons now given.

As already noted, plutons along the eastern side of Howe Sound and farther north were sampled by Symons (1973*b*). K/Ar dates from these plutons range from 90 to 117 Ma (Roddick and Woodsworth 1979) and the U/Pb zircon date of 100 Ma for the Porteau pluton indicates a mid-Cretaceous age. Symons obtained a paleopole that did not differ from *individual* mid-Cretaceous cratonic paleopoles but was strongly biased toward the late Cenozoic field: the inclinations he observed were systematically steeper than those expected for the craton (Symons 1973*b*, Fig. 2). However, his paleopole did actually differ significantly from the *mean* cratonic paleopole (Irving 1979; see also comment in Beck 1976). His results were obtained after partial demagnetization in AF fields of 10–20 mT (which would not, for example, isolate overprints carried by pyrrhotite) with no thermal demagnetization or component analysis. Hence Cenozoic magnetizations such as those we have shown to be present in the region were not identified and

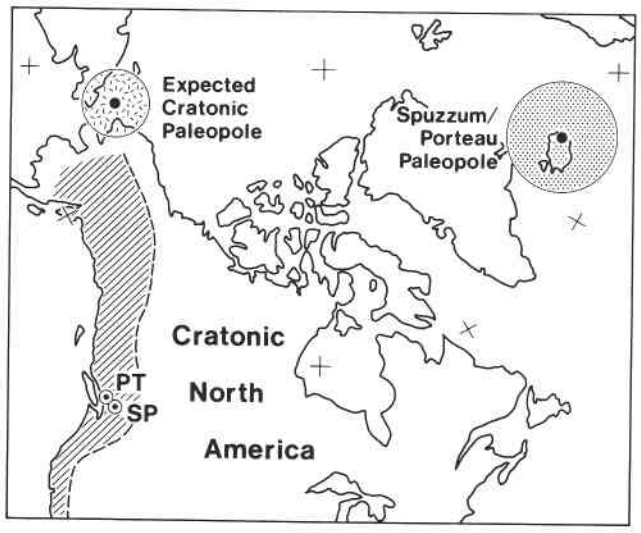


FIG. 13. Paleopole of Spuzzum (SP) and Porteau (PT) plutons compared with that expected for mid-Cretaceous of cratonic North America. Origin of mid-Cretaceous reference field given in footnote to Table 3. Errors are given for $P = 0.05$.

would not have been removed.

Moreover, Symons (1973*b*) sampled acidic granodioritic rocks including the Squamish granodiorite, which we found not to possess a stable remanent magnetization. The restriction of his work to blanket AF cleaning and also the fact that his collections include acid rocks that preserve the ancient field much less faithfully than rocks of basic and intermediate composition explain the difference between our two results.

Symons also sampled the Ecstall pluton along the Skeena River (Symons 1974) and farther south near the Hawkesbury Warp (Symons 1977*a*,*b*). The general locality is marked EP on Fig. 1. In his discussion of the paleomagnetism of the northern part of the Ecstall pluton, Symons (1974) assumed an age of about 75 ± 12 Ma for the magnetization that he observed, and, as we shall see in a moment, his assumption could be correct. Symons based his argument on a K/Ar date of 64 ± 8 Ma from biotite (from very near his site 23 or 24), and on K/Ar dates of 70 ± 4 Ma from biotite and 87 ± 15 Ma on hornblende from about 70 km southeast of his sampling traverse. Age studies subsequent to Symon's work, by Armstrong and Runkle (1979), Harrison *et al.* (1979), and others, have revealed a very complex isotopic and thermal history for the region that undoubtedly profoundly affected the magnetization. A concordant U/Pb date of 98 ± 4 Ma has been obtained from zircons of the Ecstall pluton (Woodsworth *et al.* 1983*b*) near Symon's sites 24 and 25. Harrison *et al.* (1979) obtained a Rb−Sr mineral isochron of 60 ± 3 Ma and K/Ar dates of 74 ± 1 and 70 ± 1 Ma on hornblende and biotite, respectively, from one of the samples used for the U/Pb work. They also obtained, from the same sample, a plateau date of 76 ± 1 Ma using the $^{40}Ar/^{39}Ar$ method. These results suggest that the Ecstall pluton near Symons' collecting sites 24 and 25 was emplaced at about 98 Ma and that several tens of million years later the K/Ar mineral dates were reset sometime in the latest Cretaceous or earliest Tertiary by one or more thermal events. Probable causes are movement along the Work Channel lineament and intrusion of huge early Tertiary plutons immediately to the east (Woodsworth *et al.* 1983*a*). Of the nine sites studied by Symons, sites 24 and 25 are actually the farthest from the Work Channel lineament and from the Tertiary intrusions. It is probable, therefore, that the magnetization at his other seven sites has also been reset. Hence the magnetizations observed by Symons from the Ecstall pluton were almost certainly acquired later than the original emplacement 98 Ma ago and may not, therefore, be comparable with other mid-Cretaceous results from the Cordillera. This is supported by the fact that 75% of the paleodirections are reversed (Symons 1974, Table 1, 1977*a*, Table 1) and therefore postdate the Cretaceous Normal Superchron. Whether they were acquired in the latest Cretaceous and correspond to the older reset K/Ar dates of about 75 Ma as Symons (1974) contended, or in the early Tertiary corresponding to the younger K/Ar dates and intrusion of plutons, cannot at present be established. A further complication is that movement along the Work Channel lineament was accompanied by steepening and rotation (northeastward tilting) of minor fold axes and planes to foliation (Woodsworth *et al.* 1983*a*). For this reason also the magnetization directions of the Ecstall pluton cannot at present be usefully compared with results from the other mid-Cretaceous bodies.

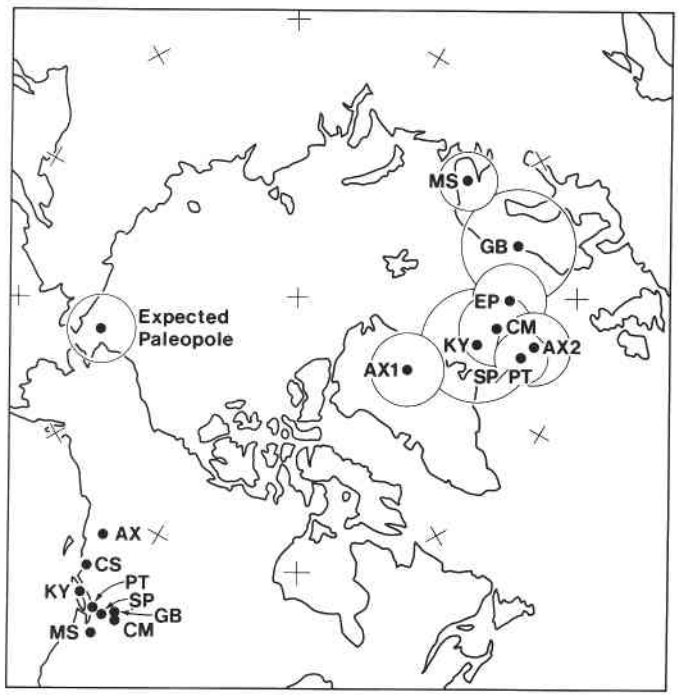


FIG. 14. Cretaceous paleopoles from the western Cordillera of Canada and the Cascade Mountains of the northwestern United States compared. The symbols are given in Table 3. In order to allow comparisons of closely situated poles, the polar errors are for $P = 0.63$, the probable error; nonintersection of error circles indicates significant difference at $P = 0.05$.

In contrast, results of Symons (1977*b*) from offshore islands farther southwest (the Captain Cove and Stephens Island plutons, CS of Table 3 and Fig. 1) probably do record the mid-Cretaceous paleofield, because the thermal history there is much less complex and they are far removed from the Work Channel lineament. Hornblende K/Ar dates range from 111 to 115 Ma and boitite K/Ar dates from 103 to 139 Ma (Roddick 1970). There were no Late Cretaceous or Tertiary thermal events to complicate the magnetic record such as occurred farther east. It is reasonable, therefore, to accept the hornblende K/Ar dates as approximately the age of the magnetizations. Paleopole CS agrees with SP/PT from the Porteau and Spuzzum bodies (Fig. 14).

The paleopoles derived from overprints in the Triassic rock

TABLE 3. Displacements and rotations calculated from mid-Cretaceous magnetizations

Symbol	Rock unit	$D°$, $I°$	$\alpha_{95}°$	$\lambda_p°$	RPD°	RR°
MS	Mount Stuart batholith	010, 45	5	27±3	31±4	−42±6
CS	Captain Cove and Stephens Island plutons	018, 58	5	39±5	26±6	−56±10
SP/PT	Spuzzum and Porteau plutons	030, 57	5	37±5	22±6	−66±8
AX1	Axelgold intrusion	025, 69	5	52±6	14±6	−66±12
AX2	Axelgold intrusion	032, 60	5	41±4	25±5	−75±10
KY	*Karmutsen Formation, overprint (Y)	025, 57	10	38±8	23±9	−60±13
CM	*Copper Mountain batholith, overprint	026, 57	7	38±6	20±6	−61±9
GBB	*Guichon batholith, overprint	020, 51	11	31±7	28±8	−57±11

NOTES: The symbols are those used in Figs. 14 and 16. Rock units are all mid-Cretaceous in age or have overprints (marked by asterisks) that are considered to be Cretaceous (see text). D, I are declination and inclination of the mean direction; α_{95} is Fisher's circle of confidence; λ_p are the paleolatitude and error. RPD and RR are the corresponding paleolatitudinal displacements and rotations (with errors) of the sampling localities relative to cratonic North America calculated using a reference pole of 67.7°N, 186.3°E, $K = 312$, $A_{95} = 3.8°$. The latter was calculated as an average of six cratonic studies (entries 39–44, age range 78–120 Ma, with a mean age of 94 Ma, Table 1, Irving and Irving 1982). The high precision ($K = 312$, standard deviation 4.6°) shows that apparent polar wander in this time interval was small or negligible. This reference pole agrees very well with that obtained for the same time interval by other workers, e.g., 68°N, 186°E by Mankinen (1978). The references are as follows: MS, Beck *et al.* (1981*b*); CS, Symons (1977*b*); SP/PT, herein; AX, Monger and Irving (1980); KY, Karmutsen overprint (Yole and Irving 1980); CM, Symons (1973*c*); GBB, Guichon batholith overprint (Symons 1976), labelled GB in Fig. 14. Errors have been calculated using the Demarest factor (Demarest 1983).

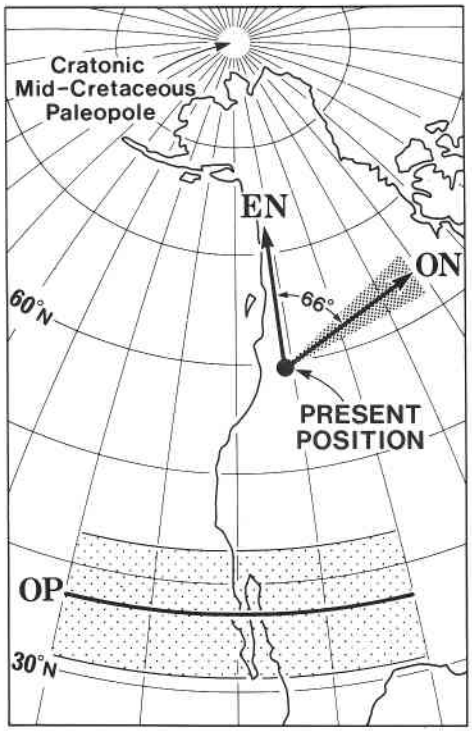


FIG. 15. Observed paleolatitude OP and rotation of the Spuzzum and Porteau plutons. The error zone ($P = 0.05$) includes errors in determining the base map (see note to Table 3). ON is observed paleonorth, EN is the direction of paleonorth expected if there had been no rotation relative to the craton.

units of Vancouver Island (Yole and Irving 1980; Schwarz *et al.* 1980) and the Quesnel block (Symons 1973*c*, 1976) are also plotted in Fig. 14 (KY, CM, and GBB). They agree with mid-Cretaceous results from the Coast Plutonic Complex and hence, we argue, are probably of that age. This group of five paleopoles (SM/PT, CS, KY, CM, GB) departs systematically from that expected from cratonic North America. It seems to us that such a departure cannot readily be explained by uniform tilting of several rock units over such a large area as that spanned by the sampling locations (about 30° of tilt to the southwest would be required). Although uniform tilting is not impossible, we regard it as more probable that large displacements of the sampling region relative to cratonic North America have occurred. In Fig. 15 and Table 3 these movements for the Spuzzum and Porteau plutons are shown. According to this reconstruction the southern part (at least) of the Coast Plutonic Complex originated about 2400 km south of its present position and subsequently rotated about 66° clockwise relative to cratonic North America. Longitude is indeterminate. Because we are dealing with events during the Cretaceous Normal Superchron when there were no reversals, there is no uncertainty in the sign of paleolatitude; the result indicates that the region was then north of the paleoequator.

The rotations indicated for Cretaceous rock units from the Canadian sector of the western Cordillera all fall in the range 56–75° clockwise, and it is notable that other rotations from Cretaceous and younger rocks from the western border of the continental United States all fall in the range of about 30–90°. For example, the rotations observed in Paleogene rocks of the Coast and Cascade ranges of Washington and Oregon range from about 30 to 90° (Bates *et al.* 1981, Fig. 6), and those in Miocene rocks of the Transverse Ranges of California vary between 70 and 80° (Luyendyk *et al.* 1980). Anticlockwise rotations, or clockwise rotations in the 90–360° range, have not been observed. All of the rotations were not produced in the same orogenic episode, because, as we have shown above (Table 2), the apparent rotation of the southern Coast Plutonic Complex of British Columbia predates the intrusion of the Chilliwack batholith and hence is older than the Miocene rocks

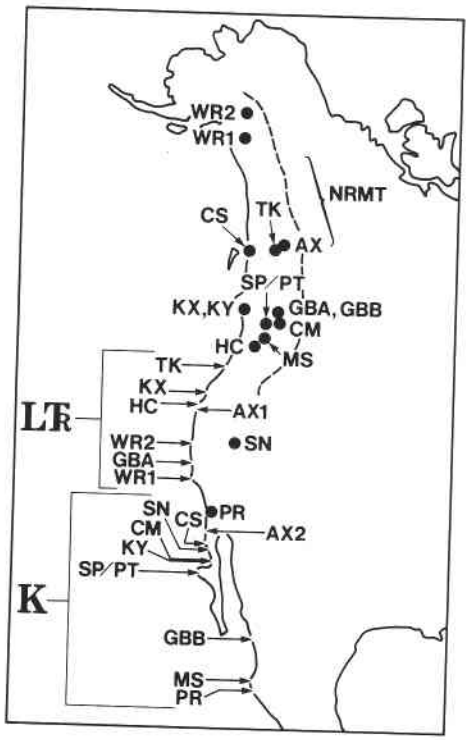


FIG. 16. Paleolatitudinal displacements of mid-Cretaceous (K) and Late Triassic (LTr) rocks compared. The northern paleolatitudes only for the latter are plotted. The dashed line is the approximate easterly limit of exotic terranes in Canada. In the north this boundary is the northern Rocky Mountain trench (NRMT) and its associated lineaments. Symbols for Cretaceous determinations are listed in Table 3. Upper Triassic determinations are as follows: GBA, probable primary magnetization from Guichon batholith, Upper Triassic (Symons 1983*b*); HC, Hell's Canyon lavas of Oregon and Idaho (Hillhouse *et al.* 1982); KX, from the primary magnetization of the Karmutsen Formation of Vancouver Island (Yole and Irving 1980); PR, Peninsular Ranges batholith (Teissere and Beck 1973) corrected for opening of Gulf of California; TK, Takla Group (Monger and Irving 1980); WR1, Nicolai Greenstone of the McCarthy quadrangle (Hillhouse 1977); WR2, Nicolai Greenstone of the Mount Hayes and Healy quadrangles (Hillhouse and Grommé 1984).

of the Transverse Ranges of California. It appears that in the Cordillera, which has undergone long-term regional shear in Cretaceous and Cenozoic times (Beck 1976), there is an upper limit to rotations of about 90°. This indicates that rotations up to this amount occur readily at a particular place, but beyond that they are not energetically favoured and strain is taken up at another place in the orogen, which becomes a mosaic of rotated blocks. Rotations greater than 90° are observed in pre-Cretaceous rocks of the Cordillera (Monger and Irving 1980), and these could be the resultant of regional shearing during earlier orogenic episodes added to that in Cretaceous and Cenozoic times.

Two other results (Axelgold layered intrusion and Mount Stuart batholith) differ significantly from the main group, although they fall in the same general region. Paleopole AX1 for Axelgold, calculated by assuming that the compositional layering was horizontal at the time the magnetization was acquired, falls in Greenland, far from its expected place (Fig. 14). Axelgold was intruded into Cache Creek Terrane just east of the Stikine block (Fig. 1). If, however, it is assumed that the present is also the paleohorizontal (i.e., either the body was intruded in its present attitude or was tilted and magnetized later during uplift) and the calculations are repeated, then the paleopole (AX2, Fig. 14) now agrees with those from the Coast Plutonic Complex. Hence the divergence of AX1 from other Cretaceous paleopoles could be caused by the incorrect choice of a reference plane, either because the dip of layering is original or because the magnetization postdates emplacement. There is also uncertainty regarding the time of emplacement of the Axelgold intrusion. Two biotites and a hornblende from a granitic xenolith yielded K/Ar ages ranging from 98 to 108 Ma. A second hornblende gave an age of 155 Ma, but is considered to "reflect the presence of excess argon" (Irvine 1975). Irvine considered a mid-Cretaceous age to be the most probable, but an Early Cretaceous or Late Jurassic age of emplacement cannot be excluded (Monger and Irving 1980). A mid-Cretaceous age for the magnetization in the Cretaceous Normal Superchron is consistent with the fact that only normal polarities were observed. It is to be expected that both normal and reversed magnetizations would be present in such a large pluton that cooled slowly, if indeed it had been magnetized during Late Jurassic or Early Cretaceous time when reversals of the geomagnetic field were frequent.

Likewise the divergence from the main group of paleopole MS for the Mount Stuart batholith (Fig. 14) could have been caused by incorrect choice of reference plane (Beck and Noson 1972; Beck *et al.* 1981*b*). The tilting required to bring MS into agreement with the paleopole from the Spuzzum–Porteau plutons is 17 ± 7° at 160°, an amount not inconsistent with the geological evidence (R. Tabor *in* Beck *et al.* (1981*b*)).

It is important in the study of the paleomagnetism of intrusions to recognise the problems that can arise from uncertainties in tilt and in the age of magnetization, and we do not wish to give the impression that we consider that the tilt versus translation problem has been conclusively resolved. Further observations are necessary to study the regional consistency of magnetizations.

Tectonic implications

In Fig. 16 the results of Fig. 14 are converted into paleolatitudes and plotted along the modern coast of North America. Although the configuration of the edge of the continent would then have been very different, this is a convenient way of expressing latitudinal displacements, since the present coastline runs essentially in the direction of Cretaceous paleonorth (Fig. 15). All magnetizations are normal and can be ascribed to the Cretaceous Normal Superchron. Consequently, all paleolatitudes are northern. It appears that during mid-Cretaceous time the southern Coast Plutonic Complex, the Cascades, Vancouver Island, and probably also the Quesnel block and Stikinia were situated in a latitude corresponding to what is today northern Mexico, and they have, since then, shared a common history of northern motion and clockwise rotation. Concordance of mid-Tertiary Cordilleran results with results from the craton (Table 2, entries 7, 8, and 9) indicate that the motion was essentially completed by then. This study clarifies the ambiguities regarding the paleomagnetism of the Coast Plutonic Complex reviewed in our Introduction and supports, for this particular complex, the general prediction of Beck *et al.* (1981*a*) that the Cretaceous batholiths of the western edge of North America were formed to the south of their present position.

It must be emphasized that longitude is indeterminate and that the sampling region could have been situated farther west, out in the Pacific. However, it is unlikely to have been far out

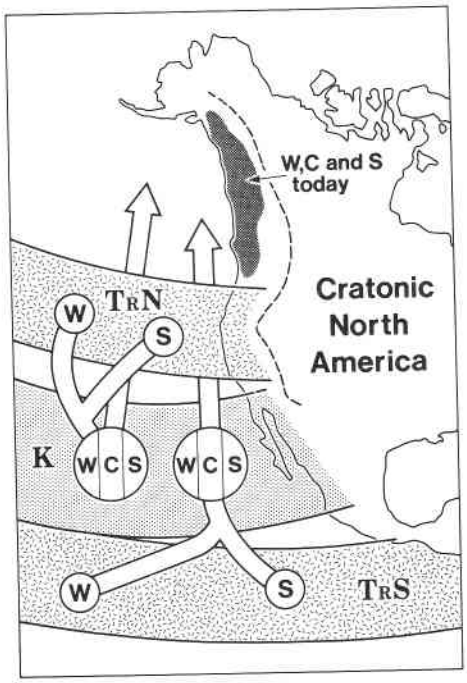


FIG. 17. Latitudinal motions of western Cordillera relative to cratonic North America. Paleolatitudinal belts are drawn to signify the paleomagnetic indeterminary of longitude. The following are symbolic representations: W, Wrangellia; C, Coast Plutonic Complex; S, Stikinia; TRS, Triassic belt southern option; TRN, Triassic belt northern option; K, Cretaceous belt.

in the Pacific because there is no evidence of there ever having been a Cretaceous ocean to the east of the sampling region and the craton, although it is conceivable that such evidence did formerly exist and was destroyed by the process of translation itself.

These estimates place the southern Coast Plutonic Complex south of where the Sierra Nevada is today and raise the question, where was the Sierra Nevada at that time? Early work by Grommé and Merril (1965) and Grommé *et al.* (1967) on plutons from the Sierra Nevada gave results that were generally consistent with those from cratonic North America. Recent detailed demagnetization studies by Frei *et al.* (1984) of three plutons from the central Sierra Nevada showed a displacement of 8 ± 10° from the south. Although the displacement is not significant at the 95% probability level, there is nevertheless a bias toward shallow inclinations, and a small poleward displacement of lower probability is indicated. These Sierra Nevadan rocks have yielded two K/Ar biotite ages of 85 Ma and K/Ar hornblende ages of 84, 92, 94, and 96 Ma, so their magnetization is probably comparable in age to the *A1* magnetization of the Spuzzum and Porteau plutons. The best estimate of 8° for this displacement places the Sierra Nevada in a paleolatitude (inclination 58 ± 3°, paleolatitude 39 ± 3°N) statistically identical to that which we have found for the southern Coast Plutonic Complex (37 ± 5°N, Table 3). Hence, the two could have been much closer together at the time of their emplacement than at present.

Results are also available from the Cretaceous Peninsular Ranges batholith (southern California and Baja California) yielding a paleolatitude of 30 ± 4° (Teissere and Beck 1973), which would place it just south of what is today the tip of Baja California. Judging from the extensive dating studies of the Peninsular Ranges batholith (Krummenacher *et al.* 1975), it is not unreasonable to suppose that the age of the magnetization is between 90 and 100 Ma. Allowing for about 4° of subsequent transport in a (Cretaceous) paleonorth direction caused by opening of the Gulf of California, an original paleolatitude of 26 ± 4° relative to cratonic North America is obtained, indicating that the Peninsular Ranges batholith was formed about 10° farther south than the Sierra Nevada and the southern Coast Plutonic Complex (Fig. 15).

The agreement between our data and those from the Sierra Nevada is striking. A similar effect is present in results from Upper Triassic rocks of the elements of Wrangellia. The present latitudinal spread of the Wrangell block, Vancouver Island, and Hills Canyon Idaho (WR, KX, and HC, respectively, of Fig. 16) is roughly three times the spread of their Late Triassic paleolatitudes. Mid-Cretaceous plutonic complexes of the western Cordillera of North America may, like Wrangellia, have originated in close proximity and have been tectonically separated and smeared out along the western margin of the continent during their northerly movements in Late Cretaceous and early Tertiary times.

Also plotted in Fig. 16 are the latitudinal displacements determined from Upper Triassic rocks of Stikinia and Quesnellia. Because the geomagnetic field reversed repeatedly during the Late Triassic Epoch, it is not known whether the paleolatitudes determined from these rocks are north or south of the paleoequator. In Fig. 16 they are assumed to be northerly, and all fall opposite the coast of central or northern California and Oregon. They are farther north than the paleolatitude determinations from the much younger Cretaceous rocks.

In Fig. 17 broad latitudinal belts are drawn to encompass Late Triassic and mid-Cretaceous results. For the reasons given above only a northerly belt is shown for the mid-Cretaceous, but both northerly and southerly options are possible for Upper Triassic rocks. Hence there are many possible paleolatitudinal trajectories. The simplest is that there has been a continuous northerly motion of all blocks of the western Canadian Cordillera from Late Triassic to mid-Tertiary times. The next simplest is that there was southerly motion of all blocks between Late Triassic and mid-Cretaceous times, followed by a northerly motion in post-mid-Cretaceous time. It is also possible that the blocks were separated in the early Mesozoic, Wrangellia could have followed the first, and Stikinia and Quesnellia (separately or together) the second trajectory, or vice versa.

General conclusions

At least four general conclusions can be drawn at this time, despite the ambiguities that still exist. These conclusions are based on the displacement interpretation of the paleomagnetic data; the alternative interpretation of tilting, although it cannot be discounted, seems to us less likely.

(1) A composite block made up of Wrangellia, the Coast Plutonic Complex, the Cascade Terrane, Stikinia, and perhaps Quesnellia probably underwent northerly displacement of about 2400 km and clockwise rotation relative to North America after the mid-Cretaceous about 90 Ma ago and before mid-Tertiary time. The amount of rotation was in the range 40–70° (Table 3) and probably varied in different parts of the composite block. The amount of displacement is greater than the minimum of 1000 km of dextral motion established by Gabrielse (1984) along the northern Rock Mountain Trench and related lineaments east of Stikinia in British Columbia and the Yukon (NRMT of Fig. 16). While his is a minimum estimate, the large difference nevertheless raises the possibility

that all of the strain did not occur east of the displaced terranes and a substantial proportion may have occurred within the terranes themselves. It is noteworthy that the paleomagnetic evidence (paleolatitudes less dispersed than present latitudes) indicates that during the northward movement the composite block was fragmented and "smeared" out along the western edge of cratonic North America. It is during this attenuation process (during Late Cretaceous or early Tertiary time) that large and variable rotations most probably occurred.

(2) Total displacements since the Late Triassic are not, as the discussion of Monger and Irving (1980) implied, the same as those since the mid-Cretaceous, showing that relative motions between these blocks and North America occurred during the interval Late Triassic to mid-Cretaceous.

(3) A general northerly motion relative to North America of all the tectonic elements of the composite block relative to North America between Late Triassic and mid-Tertiary times is consistent with the data, but so are other more complex trajectories.

(4) Fragments of at least two now widely separated mid-Cretaceous plutonic complexes of western North America (southern Coast Plutonic Complex of British Columbia and the Sierra Nevada) may have been close together when they were formed.

Acknowledgments

We are grateful to R. L. Armstrong who brought to the attention of one of us (E.I.) the fact that the Spuzzum batholith would be suitable for paleomagnetic work because of the considerable amount of geochronological information that had been obtained from it in his laboratory. We are grateful also for critical reviews by the journal referees and by R. L. Armstrong, M. J. Brandon, and J. L. Roy.

ARMSTRONG, R. L., and RUNKLE, D. 1979. Rb–Sr geochrometry of the Ecstall, Kitkiata, and Quottoon plutons and their country rocks, Prince Rupert region, Coast Plutonic Complex, British Columbia. Canadian Journal of Earth Sciences, **16**, pp. 387–399.

BARTHOLOMEW, P. R. 1979. Geology and metamorphism of the Yale Creek area, B.C. M.Sc. thesis, The University of British Columbia, Vancouver, B.C., 105 p.

BATES, G. R., BECK, M. E., and BURMESTER, R. F. 1981.Tectonic rotations in the Cascade Range of southern Washington. Geology, **9**, pp. 184–189.

BECK, M. E. 1976. Discordant paleomagnetic pole positions as evidence of regional shear in the western Cordillera of North America. American Journal of Science, **276**, pp. 694–712.

BECK, M. J., and NOSON, L. 1972. Anomalous palaeolatitude in Cretaceous granitic rocks. Nature, **235**, pp. 11–13.

BECK, M. E., BURMESTER, R. F., ENGEBRETSON, D. C., and SCHOONOVER, R. 1981*a*. Northward translation of Mesozoic batholiths of western North America: paleomagnetic evidence and tectonic significance. Geofisica Internacional, **20**, pp. 144–162.

BECK, M. E., BURMESTER, R. F., and SCHOONOVER, R. 1981*b*. Paleomagnetism and tectonics of Cretaceous Mt. Stuart batholith of Washington: translation or tilt? Earth and Planetary Science Letters, **56**, pp. 336–342.

——— 1982. Tertiary paleomagnetism of the North Cascade Range, Washington. Geophysical Research Letters, **9**, pp. 515–518.

BESKE, M. J., BECK, M. E., and NOSON, L. 1973. Paleomagnetism of the Miocene Grotto and Snoqualmie batholiths, central Cascades, Wash. Journal of Geophysical Research, **78**, 2601–2608.

DEMAREST, H. H. 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. Journal of Geophysical Research, **88**, pp. 4321–4328.

EVANS, M. E., and MCELHINNY, M. W. 1969. An investigation of the origin of stable remanence in magnetite-bearing igneous rocks. Journal of Geomagnetism and Geoelectricity, **21**, pp. 757–773.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**, pp. 295–305.

FREI, L.S., MAGILL, J. R., and COX, A. 1984. Paleomagnetic results from the central Sierra Nevada: constraints on reconstructions of the western United States. Tectonics, **3**, pp. 157–177.

GABITES, J. E. 1983. Geology and geochronology east of Harrison Lake, British Columbia. Geological Association of Canada, Annual Meeting, Victoria, B.C., Program with Abstracts, **8**, p. A25.

GABRIELSE, H. 1985. Major dextral transcurrent displacement along the northern Rocky Mountain Trench and related lineaments in north central British Columbia. Geological Society of America Bulletin, **96**, pp. 1–4.

GROMMÉ, C. S., and MERRIL, R. T. 1965. Paleomagnetism of Late Cretaceous granitic plutons in the Sierra Nevada, California—further results. Journal of Geophysical Research, **70**, pp. 3407–4320.

GROMMÉ, C. S., MERRIL, R. T., and VERHOOGEN, J. 1967. Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California, and its significance for poles wandering and continental drift. Journal of Geophysical Research, **72**, pp. 5661–5684.

HARGRAVES, R. B., and YOUNG, W. M. 1969. Source of stable remanent magnetization in Lambertville diabase. American Journal of Science, **267**, pp. 1161–1177.

HARLAND, W. B., COX, A. V., LLEWELLYN, P. G., PICKTON, C. A. G., SMITH, A. G., and WALTERS, R. 1982. A geologic time scale. Cambridge University Press, Cambridge, U.K., pp. 1–131.

HARRISON, T. M., ARMSTRONG, R. L., NAESER, C. W., and HARAKAL, J. E. 1979. Geochronology and thermal history of the Coast Plutonic Complex, near Prince Rupert, British Columbia. Canadian Journal of Earth Sciences, **16**, pp. 400–410.

HILLHOUSE, J. W. 1977. Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy quadrangle, Alaska. Canadian Journal of Earth Sciences, **14**, pp. 2578–2592.

HILLHOUSE, J. W., and GROMMÉ, C. S. 1984. Northward displacement of Wrangellia: paleomagnetic evidence from Alaska, British Columbia, Oregon and Idaho. Journal of Geophysical Research. (In press.)

HILLHOUSE, J. W., GROMMÉ, C. S., and VALLIER, T. L. 1982. Paleomagnetism and Mesozoic tectonics of the Seven Devils volcanic arc in northeastern Oregon. Journal of Geophysical Research, **87**, pp. 3777–3794.

IRVINE, T. N. 1975. Axelgold layered gabbro intrusion, McConnell Creek map-area, British Columbia. *In* Report of activities, part B. Geological Survey of Canada, Paper 75-1B, pp. 81–88.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terranes. Canadian Journal of Earth Sciences, **16**, pp. 669–694.

IRVING, E., and IRVING, G. A. 1982. Apparent polar wander paths, Carboniferous through Cenozoic and the assembly of Gondwana. Geophysical Surveys, **5**, pp. 141–188.

IRVING, E., and YOLE, R. W. 1972. Paleomagnetism and the kinematic history of mafic and ultramafic rocks in fold mountain belts. Earth Physics Branch, Energy, Mines and Resources Canada, Publication 42, pp. 87–92.

KRUMMENACHER, D., GASTIL, G., BUSHEE, J., and DOUPONT, J. 1975. K–Ar apparent ages, Peninsular Ranges batholith, southern California and Baja California. Geological Society of America Bulletin, **86**, pp. 760–768.

LUYENDYK, B. P., KAMMERLING, M. J., and TERRES, R. 1980. Geometric model for Neogene crustal rotations in southern California. Geological Society of America Bulletin, **91**, pp. 211–217.

MANKINEN, E. A. 1978. Paleomagnetic evidence for a Late Cretaceous deformation of the Great Valley Sequence, Sacramento Valley, California. United States Geological Survey, Journal of Research, **6**, pp. 383–390.

MCLEOD, J. A., VINING, M., and MCTAGGART, K. C. 1976. Note on the age of the Giant Mascot ultramafic body, near Hope, B.C. Canadian Journal of Earth Sciences, **13**, pp. 1152–1154.

MCTAGGART, K. C., and THOMPSON, R. M. 1967. Geology of part of the northern Cascades in southern British Columbia. Canadian Journal of Earth Sciences, **4**, pp. 1199–1228.

MONGER, J. W. H. 1970. Hope map-area, west half (92 H W1/2), British Columbia. Geological Survey of Canada, Paper 69–47, 75 p.

MONGER, J. W. H., and IRVING, E. 1980. Northward displacement of north-central British Columbia. Nature, **285**, pp. 289–293.

RICHARDS, T. A., and MCTAGGART, K. C. 1976. Granitic rocks of the southern Coast Plutonic Complex and northern Cascades of British Columbia. Geological Society of America Bulletin, **87**, pp. 935–953.

RODDICK, J. A. 1970. Douglas Channel – Hecate Strait map-area, British Columbia. Geological Survey of Canada, Paper 70-41, 56 p.

RODDICK, J. A., and WOODSWORTH, G. J. 1979. Geology of Vancouver west half and mainland part of Albernia. Geological Survey of Canada, Open File F6, p. 611.

SCHWARZ, E. J., MULLER, J. E., and CLARK, K. R. 1980. Paleomagnetism of the Karmutsen basalt from southeast Vancouver Island. Canadian Journal of Earth Sciences, **17**, pp. 389–399.

SYMONS, D. T. A. 1973*a*. Paleomagnetic results from the Tertiary Mount Barr and Hope plutonic complexes, British Columbia. Geological Survey of Canada, Paper 73-19, pp. 1–10.

——— 1973*b*. Concordant Cretaceous paleolatitudes from felsic plutons in the Canadian Cordillera. Nature, **241**, pp. 59–61.

——— 1973*c*. Unit correlations and tectonic rotation from paleomagnetism of the Triassic Copper Mountain intrusions, British Columbia. Geological Survey of Canada, Paper 73-19, pp. 11–28.

——— 1974. Tectonic implications of paleomagnetic results from plutons near Prince Rupert, British Columbia. Journal of Geophysical Research, **79**, pp. 2690–2697.

——— 1976. Paleomagnetism of the Triassic Guichon batholith and rotation in the interior plateau, British Columbia. Canadian Journal of Earth Sciences, **8**, pp. 1388–1396.

——— 1977*a*. Geotectonics of Cretaceous and Eocene plutons in British Columbia: a paleomagnetic fold test. Canadian Journal of Earth Sciences, **14**, pp. 1246–1262.

——— 1977*b*. Paleomagnetism of Mesozoic plutons in the westernmost Coast Plutonic Complex of British Columbia. Canadian Journal of Earth Sciences, **14**, pp. 2127–2139.

——— 1983*a*. Further paleomagnetic results from the Jurassic Topley Intrusives in the Stikinia sub-terrane of British Columbia. Geophysical Research Letters, **10**, pp. 1065–1068.

——— 1983*b*. New paleomagnetic data for the Triassic Guichon batholith and their bearing on Terrane I tectonics. Canadian Journal of Earth Sciences, **20**, pp. 1340–1343.

TEISSERE, R. F., and BECK, M. E. 1973. Divergent Cretaceous paleomagnetic pole position for the southern California batholith. Earth and Planetary Science Letters, **18**, pp. 296–300.

VINING, M. R. 1977. The Spuzzum pluton northwest of Hope, B. C. M.Sc. thesis, The University of British Columbia, Vancouver, B.C., 147 p.

WOODSWORTH, G. J., and TIPPER, H. 1980. Stratigraphic framework of the Coast Plutonic Complex, western B.C. Geological Association of Canada, Cordilleran Section, Annual Meeting 1980, Programme and Abstracts, pp. 32–34.

WOODSWORTH, G. J., CRAWFORD, M. L., and HOLLISTER, L. S. 1983*a*. Metamorphism and structure of the Coast Plutonic Complex and adjacent belts, Prince Rupert and Terrace areas, British Columbia. Geological Association of Canada, Annual Meeting 1983, Field Guide No. 14, 62 p.

WOODSWORTH, G. J., LOVERIDGE, W. D., PARRISH, R. R., and SULLIVAN, R. W. 1983*b*. Uranium–lead dates from the Central Gneiss Complex and Ecstall pluton, Prince Rupert map-area, British Columbia. Canadian Journal of Earth Sciences, **20**, pp. 1475–1483.

YOLE, R. W., and IRVING, E. 1980. Displacement of Vancouver Island: paleomagnetic evidence from the Karmutsen Formation. Canadian Journal of Earth Sciences, **17**, pp. 1210–1288.